

Kruskal's Algorithm:

1. Definition

Kruskal's algorithm is a **greedy algorithm** used to find the **Minimum Spanning Tree (MST)** of a connected, weighted, undirected graph.

It selects the edges with the **smallest weight first** while making sure that **no cycle is formed**.

ALGORITHM:

1. Start with all vertices separate.
2. Sort all edges in **increasing order of weight**.
3. Select the edge with the smallest weight.
4. Check whether adding the edge creates a cycle.
5. if it does **not** create a cycle, add it to the MST.
6. If it creates a cycle, reject that edge.
7. Repeat until **$n - 1$ edges** are selected.
8. Calculate the total cost of the selected edges.

Time Complexity:

Case	Time Complexity
------	-----------------

Best Case	$O(E \log E)$
-----------	---------------

Average Case	$O(E \log E)$
--------------	---------------

Worst Case	$O(E \log E)$
------------	---------------

Space Complexity:

Space complexity: $O(V + E)$

OUTPUT:

```
Enter number of vertices: 4
Enter number of edges: 5
Enter edges (u v weight):
1 2 1
1 3 2
1 4 4
2 4 3
3 4 5

Minimum Spanning Tree:
1 - 2 = 1
1 - 3 = 2
2 - 4 = 3
Total cost = 6

...Program finished with exit code 0
Press ENTER to exit console.
```

Conclusion :

Kruskal's algorithm is a simple and efficient greedy algorithm for finding the Minimum Spanning Tree. It selects the smallest-weight edges and avoids cycles. It is especially useful when the graph has a relatively small number of edges.